

CIS 5800 Exam 1, Spring 2025

Name:

Student ID:

Max time: 90 minutes

Please write your name and ID on each page

Formulae Sheet

Please note: A point with pixel (uncalibrated) coordinates $(u, v) \in \mathbb{R}^2$ is embedded to a point $(u, v, 1) \in \mathbb{P}^2$. When we write a point $(u, v, w) \in \mathbb{P}^2$, (u, v) is not in pixels, but $(u/w, v/w) \in \mathbb{R}^2$ is in pixels.

The projection equation from a point (X_w, Y_w, Z_w, W_w) (in projective space P^3) with respect to a world coordinate system to a point (u, v, w) (in projective space P^2) on the image is given by:

$$\lambda \begin{pmatrix} u \\ v \\ w \end{pmatrix} = K \begin{pmatrix} R & t \end{pmatrix} \begin{pmatrix} X_w \\ Y_w \\ Z_w \\ W_w \end{pmatrix}$$

If the projection is NOT at infinity in the image plane, its pixel coordinates read $(u/w, v/w)$.

If the point is not at infinity in the world, its metric coordinates read $(X_w/W_w, Y_w/W_w, Z_w/W_w)$.

Cross Ratio is defined as:

$$\text{Cross Ratio}\{A, B; C, D\} = \frac{AC}{AD} : \frac{BC}{BD}$$

Rotation matrixes around coordinate system axes:

$$R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix} \quad R_y = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix} \quad R_z = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Cross Product and Skew matrix:

$$a \times b = \hat{a}b \quad \text{where} \quad \hat{a} = \begin{pmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{pmatrix}$$

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Problem 1: (10pts)

1.1 - Basics (3pts)



Figure 1: Hallway intersection at Penn

Figure 1 depicts a hallway intersection in an engineering building at Penn. Assume that:

(i) The photo is free of radial distortion. (ii) The vertical vanishing point is at infinity (Vertical vanishing points are the projections of points at infinity where parallel lines orthogonal to the ground plane appear to converge).

Using a pen, sketch the horizontal vanishing points and the horizon line for the ground plane directly on the image. (Horizontal vanishing points are the projections of points at infinity where parallel lines on the ground plane appear to converge)

(Note: Please do not use pencils, as they may not be scanned clearly.)

1.2 - The Use of Vanishing Points: (3pts)

Answer the following questions with **True** or **False**

- All the vanishing points you can draw in the photo (i.e., excluding those at infinity) are on the same line.
- In the photo, the vertical vanishing point at infinity v_{vert} can be expressed by $(x, y, 1)$ in the homogeneous coordinates.
- With the horizon line you drew in the photo, you can determine the camera orientation along the principal axis w.r.t the ground.

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1.3 - Cross Ratio (4pts)



Figure 2: Hallway intersection at Penn, with vanishing point V

Going back to the hallway intersection in Figure 1, we define the corresponding points illustrated in Figure 2:

- The vanishing point as V.
- The point in front of the nearest classroom door is C.
- Two corners, A and B, represent the thickness of a structure between two buildings.

A, B, C, V lies on a straight line in the image. Given the image measurements $AB = 0.6$, $BC = 1.2$, $CV = 1.2$, and knowing that the actual distance $A_W C_W = 10m$, calculate the thickness $A_W B_W = \omega$ of the structure between the buildings.

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Problem 2 - Transformations (10pts)

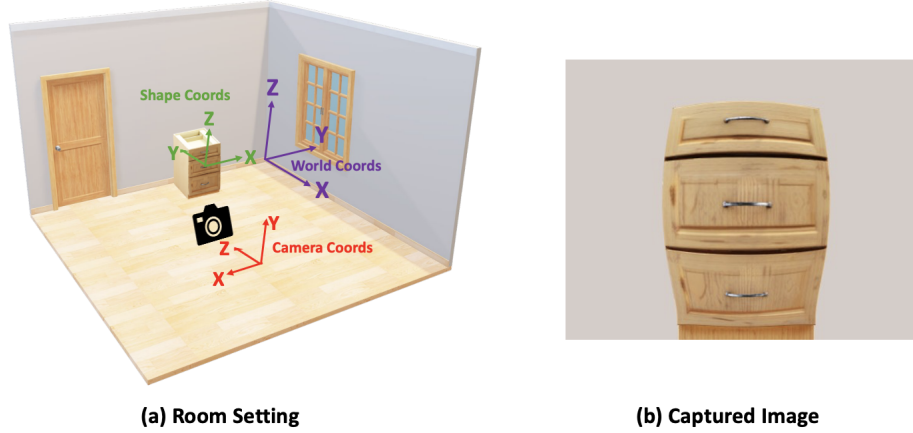


Figure 3: Augmented Reality for the cabinet in the room

As shown in Fig. 3 (a), we are trying to take a picture of a cabinet placed in a room, where we have three coordinate frames. Any point X , can be represented by its coordinates in the world frame X_w , camera frame X_c , and shape frame X_s . Assume we know (R_{cw}, T_{cw}) and (R_{sw}, T_{sw}) such that:

$$\begin{aligned}X_w &= R_{cw}X_c + T_{cw} \\X_w &= R_{sw}X_s + T_{sw}\end{aligned}$$

(a) (4pts) What are the expressions of the rotation matrix R_{sc} and translation vector T_{sc} that projects a point from shape coordinates to camera coordinates (i.e. $X_c = R_{sc}X_s + T_{sc}$)? Your answer should be expressed in terms of (R_{cw}, T_{cw}) and (R_{sw}, T_{sw})

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(b) (3 pts) Given the camera intrinsics matrix K , express the projection of the origin of the shape coordinate frame in pixel coordinates using (R_{cw}, T_{cw}) , (R_{sw}, T_{sw}) , and K .

(c) (3pts) Assume that the camera intrinsics matrix is expressed as $\begin{pmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Is it possible for this camera to capture the image shown in Fig. 3 (b)? Why or why not?

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Problem 3 - Homographies (10pts)

Given four world plane coordinates and their corresponding pixel coordinates, as well as the camera's intrinsic matrix K , please answer the following questions:

- (1) (6 pts) Calculate a projective transformation (i.e., Homography) H that maps the world plane points $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, and $(1, 1, 1)$ to the pixels on the image plane $(2, 0, 1)$, $(0, 1, 1)$, $(3, 0, 1)$, and $(2, 1, 1)$ using the 4-point algorithm.

- (2) Recall the projection from world plane coordinates to the pixel coordinates:

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} \sim K \begin{pmatrix} r_1 & r_2 & r_3 & T \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ W \end{pmatrix}$$

Assume that all points on the world plane lie on $Z = 0$.

Then the transformation becomes:

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} \sim K \begin{pmatrix} r_1 & r_2 & T \end{pmatrix} \begin{pmatrix} X \\ Y \\ W \end{pmatrix}$$

In class, we learned that $K \begin{pmatrix} r_1 & r_2 & T \end{pmatrix} = H$, or in other words:

$$K^{-1}H = \begin{pmatrix} r_1 & r_2 & T \end{pmatrix}, \quad \text{and} \quad r_3 = r_1 \times r_2$$

- (a) (2 pts) Can we use the first two columns of $K^{-1}H$ as the first two columns of the camera rotation matrix? Why or why not?

- (b) (2 pts) Fill in the blank to complete the optimization objective for recovering the camera extrinsic parameters. Suppose $H' = K^{-1}H$, and denote the columns of H' as $(h'_1 \ h'_2 \ h'_3)$:

$$\|(h'_1 \ h'_2 \ h'_3) - \text{_____}\|_F$$

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Problem 4: (10pts) PnP solves the camera system that consists of:

- A **rotation matrix** R (3×3 matrix)
- A **translation vector** T (3×1 vector)

The minimum number of point correspondences required in PnP is 3, which is also called P3P.

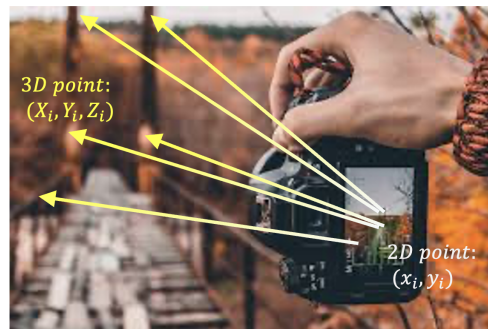


Figure 4: 2D-3D correspondences in the camera system.

(a) (3 pts) Why will PnP only require 3 point correspondences? (Hint: consider the degrees of freedom and how many constraints each correspondence provides.)

(b) (2 pts) However, to formulate PnP in a more general form, we do not enforce the orthogonality constraint on the rotation matrix at the initial stage of computing R . As a result, R is treated as a full 3×3 matrix with 9 independent variables. In this case, what is the **minimum number of point correspondences** (as shown in Figure 4) required to solve this general form of PnP?

(c) (5 pts) Given n pairs of correspondences $\{(x_i, y_i, X_i, Y_i, Z_i)\}_{i=1}^n$, such that:

$$\lambda_i \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = R \begin{bmatrix} X_i \\ Y_i \\ Z_i \end{bmatrix} + T$$

where λ_i is a scaling factor. We can solve the PnP problem as a homogeneous linear system $\mathbf{A}_{2n \times 12} \mathbf{x} = 0$, where:

$$\mathbf{x} = (r_{11}, r_{12}, r_{13}, r_{21}, r_{22}, r_{23}, r_{31}, r_{32}, r_{33}, T_1, T_2, T_3)^T$$

Derive the explicit form of **two rows** of the matrix $\mathbf{A}$ obtained from one pair of point correspondences. Express your answer as a 2×12 matrix in terms of $(x_i, y_i, X_i, Y_i, Z_i)$. (Hint: Consider how to eliminate the scale factor λ_i .)

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Problem 5: (10 pts)

You are given a set of non-collinear calibrated points $\{p_i\}_{i=1}^8, \{q_i\}_{i=1}^8$ from camera frames A and B , respectively. The points satisfy the epipolar constraint $q_i^T(\hat{T}R)p_i = 0$, where $T \in \mathbb{R}^3$ and $R \in SO(3)$ are as indicated in the diagram:

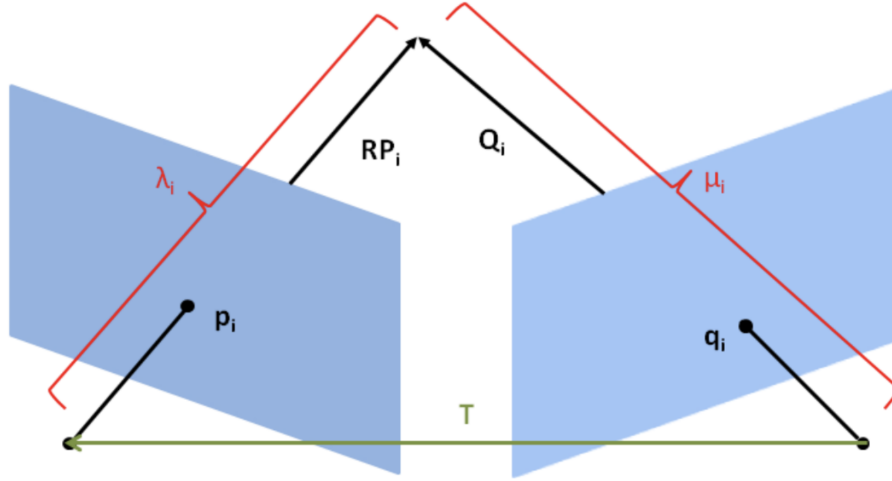


Figure 5: $R(\lambda p) + T = \mu q$

(a) (4pts). Indicate on the graph above the position of the epipole on each frame. Draw the epipolar lines associated with points p_i and q_i .

(b) (1pts). For each pair of points (q, p) , we can reformulate the epipolar constraint as follows:

$$0 = q^T \begin{bmatrix} e_1 & e_2 & e_3 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad (1)$$

$$= q^T [p_x e_1 + p_y e_2 + p_z e_3] \quad (2)$$

$$= \begin{bmatrix} p_x q^T & p_y q^T & p_z q^T \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} \quad (3)$$

where e_1, e_2, e_3 are the columns of the Essential matrix E . Is Eq. (3) a linear equation? (Yes/No)

(c) (2pts). Given the following SVD decomposition of E : $E = U \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} V^T$, if E is a valid essential matrix, what does the relationship between σ_1 and σ_2 indicate and what should the value of σ_3 be?

(d) (3pts). After obtaining a valid Essential matrix E , we may obtain four analytic solutions for (R, T) , briefly explain how you can determine the correct (R, T) .

Scratch Paper

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